Database Design Document (DDD)
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1. Overview

This document defines the logical database design, entity relationships, normalization approach, primary and foreign keys, indexing strategy, and business rules for the Supabase PostgreSQL backend.

2. Database Architecture

Supabase PostgreSQL is the primary datastore. UUIDs are used for primary keys. All business entities include created_at and updated_at timestamps. Row Level Security (RLS) protects administrator data.

Table: profiles

Column	Type	Purpose
id	UUID	Admin user ID
full_name	TEXT	Administrator name
email	TEXT	Login email
role	TEXT	admin/super_admin

Business Rules

This entity participates in foreign-key relationships, supports indexed lookups, and follows normalization principles to avoid redundancy while maintaining data integrity.

Table: categories

Column	Type	Purpose
id	UUID	Primary Key
name	TEXT	Category name
slug	TEXT	SEO slug

Business Rules

This entity participates in foreign-key relationships, supports indexed lookups, and follows normalization principles to avoid redundancy while maintaining data integrity.

Table: products

Column	Type	Purpose
id	UUID	Primary Key
category_id	UUID	FK -> categories
name	TEXT	Product name
sku	TEXT	Unique SKU
price	NUMERIC	Selling price
stock	INTEGER	Available stock
active	BOOLEAN	Visibility

Business Rules

This entity participates in foreign-key relationships, supports indexed lookups, and follows normalization principles to avoid redundancy while maintaining data integrity.

Table: orders

Column	Type	Purpose
id	UUID	Primary Key
order_number	TEXT	Unique order ID
session_id	UUID	Guest session
total	NUMERIC	Order total
order_status	TEXT	Lifecycle state

Business Rules

This entity participates in foreign-key relationships, supports indexed lookups, and follows normalization principles to avoid redundancy while maintaining data integrity.

Table: order_items

Column	Type	Purpose
order_id	UUID	FK -> orders
product_id	UUID	FK -> products
quantity	INTEGER	Units
unit_price	NUMERIC	Snapshot price

Business Rules

This entity participates in foreign-key relationships, supports indexed lookups, and follows normalization principles to avoid redundancy while maintaining data integrity.

Entity Relationships

Categories have many Products. Products have many Product Images. Guest Sessions own Carts. Carts contain Cart Items. Orders contain Order Items. Payments belong to Orders. Profiles manage Inventory Logs and Audit Logs.

Indexes

Create indexes on SKU, slug, category_id, order_number, payment identifiers, email fields, and created_at for efficient querying.

Normalization

The schema is designed in Third Normal Form (3NF) to eliminate redundancy while preserving efficient joins.

RLS Policies

Anonymous users can read active products. Only authenticated admins can modify products, inventory, categories, and orders. Payments are verified through the backend.

Future Tables

Reviews, wishlists, coupons, blogs, FAQs, loyalty points, notifications, and AI recommendations can be added without major schema changes.